words about wavefunction collapse or the imprecise notion of “large” systems belonging to the dominion of classical physics, unnerving. To a significant extent, by taking account of decoherence, researchers have rendered these vague ideas unnecessary.

However, a key issue that I skirted in the description above is that even though decoherence suppresses quantum interference and thereby coaxes weird quantum probabilities to be like their familiar classical counterparts, *each of the potential outcomes embodied in a wavefunction still vies for realization.* And so we are still left wondering how one outcome “wins” and where the many other possibilities “go” when that actually happens. When a coin is tossed, classical physics gives an answer to the analogous question. It says that if you examine the way the coin is set spinning with adequate precision, you can, in principle, *predict* whether it will land heads or tails. On closer inspection, then, precisely one outcome is determined by details you initially overlooked. The same cannot be said in quantum physics. Decoherence allows quantum probabilities to be interpreted much like classical ones, but does not provide any finer details that select one of the many possible outcomes to actually happen.

Much in the spirit of Bohr, some physicists believe that searching for such an explanation of how a single, definite outcome arises is misguided. These physicists argue that quantum mechanics, with its updating to include decoherence, is a sharply formulated theory whose predictions account for the behavior of laboratory measuring devices. And according to this view, *that* is the goal of science. To seek an explanation of *what’s really going on,* to strive for an understanding of *how a particular outcome came to be,* to hunt for a level of *reality beyond detector readings and computerprintouts* betrays an unreasonable intellectual greediness.

Many others, including me, have a different perspective. Explaining data *is* what science is about. But many physicists believe that science is also about embracing the theories data confirms and going further by using them to gain maximal insight into the nature of reality. I strongly suspect that there is much insight to be gained by pushing onward toward a complete solution of the measurement problem.